Ching-Wen Hung

Ph.D. Candidate in Computer Science
Graduate Institute of Networking and Multimedia
National Taiwan University
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Research Interests

- Human–Computer Interaction (HCI), Embodied Interaction, Haptic and Gesture Interfaces, Virtual/Augmented Reality (VR/AR), Perception and Inclusive Interactive Systems

Education

- **National Taiwan University** — **Taipei, Taiwan**
  Ph.D. in Computer Science (Direct Ph.D. Program) — *Mar. 2020 - Present*
  Advisor: Bing-Yu Chen
- **National Taiwan University** — **Taipei, Taiwan**
  Graduate Coursework in Information Management — *Sep. 2018 - Feb. 2020*
  Entered Direct Ph.D. Program in Computer Science
  Advisor: Bing-Yu Chen
- **National Central University** — **Taoyuan, Taiwan**
  Bachelor of Business Administration in Information Management — *Sep. 2014 - Jun. 2018*

Research Experiences

- **Visiting Faculty Assistant - Embodied Dynamics Laboratory** — *Oct. 2023 - Sep. 2024*
  University of Maryland, College Park, Maryland, USA
  Supervisor: Jun Nishida
- **Research Assistant - Communications and Multimedia Laboratory** — *Sep. 2018 - Present*
  National Taiwan University, Taipei, Taiwan
  Supervisor: Bing-Yu Chen

Publications

- **DilatArm: An Accessible, Low-cost Passive Exoskeleton using Dilatant Fluid for Viscosity Feedback during Spatial Movement**
  Ching-Wen Hung, Eric Chen, Hamza Shah Khan, Po-Ying Hsu, Bing-Yu Chen, Jun Nishida
  In Proceedings of the ACM Symposium on Spatial User Interaction **(SUI 2026)**, *to appear.* (oral present)
- **ElderPlay: Supporting Age-Inclusive Gameplay for Older Adults via Real-Time Gesture-to-Controller Translation**
  Ching-Wen Hung, Che-Wei Hsu, Wei-Tang Hsu, Tzu-Chin Chiu, Li Lin, Yao Cheng Lee, Ting-Wu Chang, Hsien-Hui Tang, Bing-Yu Chen, Mike Y. Chen
  In Proceedings of the ACM Designing Interactive Systems Conference **(DIS 2026)** (oral present) [accepted rate: 21%]
- **TacNote: Tactile and Audio Note-Taking for Non-Visual Access**
  Wan-Chen Lee, **Ching-Wen Hung**, Chao-Hsien Ting, Peggy Chi, Bing-Yu Chen
  In Proceedings of the ACM Symposium on User Interface Software and Technology **(UIST 2023)**. (oral present) [accepted rate: 25.1%]
- **Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions**
  Ching-Wen Hung, Ruei-Che Chang, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen
  In Proceedings of the ACM Symposium on Virtual Reality Software and Technology **(VRST 2022)**. (oral present) [accepted rate: 26.7%]

- **OsciHead: Simulating Versatile Force Feedback on an HMD by Rendering Various Types of Oscillation**
  Ching-Wen Hung, Hsin-Ruey Tsai, Chi-Chun Su, Jui-Cheng Chiu, Bing-Yu Chen
  In Proceedings of the ACM Human-Computer Interaction with Mobile Devices and Services **(MobileHCI 2022)**.
  In Proceedings of the ACM on Human-Computer Interaction **(PACMHCI 2022)**. (oral present) [accepted rate: 36%]
- **ElastOscillation: 3D Multilevel Force Feedback for Damped Oscillation on VR Controllers**
  Hsin-Ruey Tsai, **Ching-Wen Hung**, Tzu-Chun Wu, Bing-Yu Chen
  In Proceedings of the ACM Human-Computer Interaction **(CHI 2020)**. (oral present) [accepted rate: 24.31%]

Projects, Posters, Demos

- **Designing Embodied Interaction: Shaping Bodily Action, Experience, and Capability**
  Ching-Wen Hung
  In Proceedings of the ACM Symposium on User Interface Software and Technology **(UIST 2026 Doctoral Symposium)**.
- **Demonstration of DilatArm: An Accessible, Low-Cost Passive Exoskeleton Powered by Dilatant Fluid for Velocity-Dependent Force Feedback**
  Eric Chen, **Ching-Wen Hung**, Hamza Shah Khan, Po-Ying Hsu, Bing-Yu Chen, Jun Nishida
  In Proceedings of the ACM Symposium on Spatial User Interaction **(SUI 2026 Demo)**.
  Best Demo Award
- **FingerPuppet: Finger-Walking Performance-based Puppetry for Human Avatar**
  Ching-Wen Hung, Chung-Han Liang, Bing-Yu Chen
  In Proceedings of the ACM Human-Computer Interaction **(CHI 2024 Late-Breaking Work)**. [accepted rate: 33.88%]
- **Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper-Body Postures**
  Ching-Wen Hung, Ruei-Che Chang, Hong-Sheng Chen, Chung Han Liang, Liwei Chan, Bing-Yu Chen
  In Proceedings of the ACM Symposium on User Interface Software and Technology **(UIST 2022 Adjunct)**.
- **Tonopoly: A Monopoly-like Board Game with Toio**
  Chi-Huan Chiang, **Ching-Wen Hung**
  In Proceedings of the ACM Symposium on User Interface Software and Technology **(UIST 2021 Student Innovation Contest)**.
- **Demonstration of ElastOscillation: A VR Controller Providing 3D Multilevel Feedback for Damped Oscillation**
  Ching-Wen Hung, Hsin-Ruey Tsai, Tzu-Chun Wu, Bing-Yu Chen
  In Proceedings of the ACM Human-Computer Interaction **(CHI 2020 Extended Abstracts)**.

Invited Talks and Presentations

- **Embodied Interaction: Enhancing Immersive Interaction in Virtual and Physical Realities**
  Seminar Talk at National Chung Cheng University (Chiayi, Taiwan) 2024
- **PhD Research in HCI: Exploring Embodied Interaction and Funding Opportunities in Taiwan**
  Podcast (Online) 2024
- **TacNote: Tactile and Audio Note-Taking for Non-Visual Access**
  UIST 2023 (San Francisco, California, USA)
  TAICHI 2023 (Taipei, Taiwan)
- **Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions**
  VRST 2022 (Tsukuba, Japan)
  TAICHI 2022 (Hsinchu, Taiwan)

- **Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper-Body Postures**
  UIST 2022 Adjunct (Bend, Oregon, USA)
- **OsciHead: Simulating Versatile Force Feedback on an HMD by Rendering Various Types of Oscillation**
  MobileHCI 2022 (Online Virtual Conference)
  TAICHI 2021 (Online Virtual Conference)
- **ElastOscillation: 3D Multilevel Force Feedback for Damped Oscillation on VR Controllers**
  CHI 2020 (Online Virtual Conference)
  UTokyo-NTU Joint Conf 2019 (Tokyo, Japan)

Teaching Certifications

- English-Medium Instruction (EMI) Teaching Assistant Certification, National Taiwan University, 2026
- Mandarin-Medium Teaching Assistant (TA) Certification, National Taiwan University, 2020

Teaching Experiences

- **Teaching Assistant**
  - **Human-AI Interaction and Collaboration** *National Taiwan University* *Sep. 2026 – Dec. 2026*
    Instructor: Yi-Chi Liao
    * English-medium instruction (EMI) course
    * Facilitated project milestone presentations, including coordinating presentation schedules and evaluating student presentations.
    * Evaluated project milestones, including written submissions, live demos, and presentations, with emphasis on Human-AI interaction design, system functionality, and project quality.
    * Graded course assignments and supported course activities, including technical setup for LLM-based tools.
  - **Multimodal Human-Computer Interaction** *National Taiwan University* *Feb. 2026 - Jun. 2026*
    Instructor: Shan-Yuan Teng
    * English-medium instruction (EMI) course
    * Led hands-on workshops on haptic interaction and robotic arm prototyping using MeArm kits, including servo calibration, Arduino control, and system assembly
    * Designed and 3D-printed customized components to support student assignments and facilitate exploration of haptic applications
    * Mentored students in multimodal prototyping across visual, audio, and haptic modalities, providing technical guidance on Processing, Arduino, and hardware integration
    * Advised final course projects on interaction design, research methodology, and user study planning, including proposal feedback and experimental design review
    * Evaluated student mini-project expos and final projects by providing design critique, usability feedback, and research-oriented assessment across multimodal interaction assignments
  - **Game Programming** *National Taiwan University* *Sep. 2025 - Dec. 2025*
    Instructor: Ken-Yi Lee, Bing-Yu Chen
    * Mandarin-medium instruction course
    * Assisted with grading and course administration
    * Coordinated the final project demo workshop, facilitating student presentations and assessment by instructors and invited industry professionals
  - **Human Computer Interaction in Industry** *National Taiwan University* *Sep. 2022 - Dec. 2022*
    Instructor: Peggy Chi
    * English-medium instruction (EMI) course
    * Facilitated student discussions and project development for industry-oriented HCI topics
    * Provided feedback and guidance on final project presentations

- **Creative and Design Thinking** *National Taiwan University* *Mar. 2020 - Jun. 2020*
  Instructor: Bing-Yu Chen
  * Mandarin-medium instruction course for executive entrepreneurship program students (NTU EiMBA)
  * Facilitated two-day design thinking workshops
  * Assisted with team formation, activity coordination, and final presentations
- **Internet Technology in E-Commerce** *National Central University* *Sep. 2017 - Jan. 2018*
  Instructor: Kuen-Liang Su, Shi-Jen Lin
  * Mandarin-medium instruction course (In-service Master's Program)
  * Supported classroom coordination and course administration for working professionals
  * Assisted with group organization, attendance tracking, and grading records

Student Mentoring

- **Bo-Ru Chen***, *Master Student*, National Taiwan University *Jul. 2025 - Present*
- **Che-Wei Hsu***, *Undergraduate Student*, National Taiwan University *Jul. 2025 - Jun. 2026*
- **Wei-Tang Hsu***, *Undergraduate Student*, National Taiwan University *Jul. 2025 - Jun. 2026*
- **Po-Ying Hsu***, *Master Student*, National Taiwan University *Dec. 2024 - Jul. 2025*
- **Wan-Chen Lee***, *Master Student*, National Taiwan University *Dec. 2022 - Oct. 2023*
- **Min-Rui Li**, *Master Student*, National Taiwan University *Feb. 2023 - Jun. 2023*
- **Wei-Xun Huang**, *Master Student*, National Taiwan University *Feb. 2023 - Jun. 2023*
- **Hong-Sheng Chen***, *Master Student*, National Taiwan University *Nov. 2021 - Apr. 2022*
- **Chung Han Liang***, *Master Student*, National Taiwan University *Nov. 2021 - Apr. 2022*
- **Chi-Chun Su***, *Master Student*, National Taiwan University *Aug. 2020 - Aug. 2021*
- **Jui-Cheng Chiu***, *Undergraduate Student*, National Taiwan University *Aug. 2020 - Apr. 2021*

* Co-authored peer-reviewed publications.

Academic and Co-Curricular Teaching

- Programming Instructor and Executive Member, NCU Big Data and Programming Society *2016–2017*
  - Delivered programming workshops (Web, Android, Linux) for undergraduate students
  - Developed instructional materials and hands-on coding exercises
- C Programming Instructor, 21st NCU MIS Camp *2015*
  - Designed and delivered introductory C programming workshops for high school students
  - Developed course materials and interactive programming activities

Honors and Awards

- **Best Demo Award** *2026*
  Demonstration of DilatArm
  ACM Symposium on Spatial User Interaction (SUI 2026), Bari, Italy
- **Gary Marsden Travel Award** *2022*
  Selected recipient; declined in favor of ACM UIST Travel Award
  ACM SIGCHI
- **Dean's List Award** *2015 - 2017*
  Recognized for outstanding academic performance
  National Central University, Taiwan

Fellowships and Grants

- **Graduate Student International Conference Travel Grant** *2026*
  Awarded two travel grants totaling approximately USD 1,320 for ACM DIS 2026 and ACM UIST 2026
  College of Electrical Engineering and Computer Science, National Taiwan University

- **Outstanding Student Conference Travel Grant** *2026*
  Awarded approximately USD 2,138 for presenting research at ACM SUI 2026, Bari, Italy
  Foundation for the Advancement of Outstanding Scholarship, Taiwan
- **Ph.D. Students Study Abroad Program** *2023 - 2024*
  Supported one-year visiting research appointment at the Department of Computer Science, University of Maryland, College Park
  (approximately USD 30,000 in funding support)
  National Science and Technology Council, Taiwan
- **UIST Travel Award** *2022*
  Received travel grant support of USD 4,433 for presenting research at ACM UIST 2022, Bend, Oregon
  ACM UIST
- **Subsidy for Domestic Graduate Students to Attend International Academic Conferences** *2022*
  National Science and Technology Council, Taiwan
- **Fellowship for Outstanding Ph.D. Student** *2020 - 2024*
  Awarded monthly fellowship support for outstanding Ph.D. students
  Ministry of Science and Technology, Taiwan
- **Fellowship for Students of the Direct Pursuit of Ph.D. Degree** *2020*
  Awarded monthly fellowship support upon entering the Direct Ph.D. Program
  National Taiwan University, Taiwan

Leadership

- **Coordinator, Taiwan–Japan HCI Joint Workshop** *2026*
- **Session Chair, Taiwan HCI Internal Pre-CHI Workshop** *2026*
- **Project Leader, CMLab (Robin Group)** *2020 - 2023, 2026*
- **Instructor, CMLab (Robin Group) HCIxGraphics Workshop** *2020 - 2022*
- **Instructor, CMLab (Robin Group) HCI Workshop** *2019*
- **Host, TAICHI 2022 – Women in STEM Luncheon** *2022*
- **Host, TAICHI x UIST Online Meeting** *2021*

Professional Services

- **Conference Committee Roles**
  - TAICHI 2026 – Demo Chair
  - TAICHI 2025 – Demo Chair
  - TAICHI 2023 – Demo Chair
  - TAICHI 2022 – Program Chair
- **Reviewer**
  - CHI 2027
  - VRST 2026
  - TAICHI 2026 (Demo)
  - AHs 2026
  - CHI 2026 (Papers & Posters)
  - SIGGRAPH Asia 2025 (ET)
  - TAICHI 2025 (Demo)
  - CGW 2025
  - UIST 2025
  - CHI 2025
  - TAICHI 2024
  - CHI 2024 (LBW)

  - TAICHI 2023
  - CHI 2023 (LBW)
  - AHs 2023
  - TAICHI 2022
  - TAICHI 2021
- **Student Volunteer**
  - UbiComp 2021
  - CHI 2021
  - TAICHI 2019

References

- **Bing-Yu Chen**
  Distinguished Professor at Department of Computer Science and Information Engineering / Department of Information Management / Graduate Institute of Networking and Multimedia, National Taiwan University
  Email: robin@ntu.edu.tw
- **Jun Nishida**
  Assistant Professor at Department of Computer Science, University of Maryland, College Park
  Email: jun@umd.edu
- **Shan-Yuan Teng**
  Assistant Professor at Department of Computer Science & Information Engineering, National Taiwan University
  Email: tengshanyuan@csie.ntu.edu.tw
- **Mike Y. Chen**
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- **Hsien-Hui Tang**
  Professor of Department of Commercial and Industrial Design, National University of Science and Technology
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- **Liwei Chan**
  Associate Professor at Institute of Computer Science and Engineering, National Yang Ming Chiao Tung University
  Email: liweichan@cs.nycu.edu.tw